Optimizing 3D Low-Poly Asset Production Standards for Village Meow 2.5D Game Using GDLC with Expert Review

Aisyah Beningsari Mahadi, Anis Rahmi*, Nursaima Harahap, Liony Lumombo

Politeknik Negeri Batam, Jl. Ahmad Yani,Kota Batam, 29461, Indonesia
*anis@polibatam.ac.id

Abstract. This research focuses on optimizing 3D low-poly asset production standards for the Village Meow 2.5D game using a modified Game Development Life Cycle (GDLC) methodology with expert review validation. The development process was conducted during the IGDX Bootcamp 2024 program, concentrating on 3D environment asset creation with emphasis on modeling, texturing, and performance optimization. A descriptive qualitative method combined expert interviews with Unity Profiler performance testing for comprehensive evaluation. Results demonstrate that most assets meet visual and technical requirements, with an average triangle count of 15,000-20,000 triangles for medium-complexity assets. However, several assets required optimization to fully comply with low-poly criteria. The modified GDLC approach proved effective in streamlining asset creation workflow while maintaining quality standards. Expert review validation confirmed that the optimized production pipeline significantly improved asset consistency and reduced development time. This research contributes to understanding efficient 3D asset production methodologies for independent game development.

Keywords: 3D Low-Poly Assets, Game Development, GDLC Methodology, Asset Production Pipeline, Expert Review Validation, Unity Optimization

1 Introduction

The digital 3D asset production industry has experienced remarkable growth, with the global 3D digital asset market projected to expand from USD 28.3 billion in 2024 to USD 51.8 billion by 2029, representing a Compound Annual Growth Rate (CAGR) of 12.9% [1]. This growth is primarily driven by the rapid expansion of the gaming and entertainment industries, which increasingly depend on high-quality 3D assets, with the 3D gaming technology market projected to reach USD 34.5 billion in 2024 [2].

Beyond the gaming and entertainment sector expansion, significant advancements in 3D graphics technology and game engine development have contributed to this substantial growth. These technologies enable real-time rendering, complex animations, and more efficient asset creation workflows. GPU technology improvements and the

H. Widiastuti et al. (eds.), *Proceedings of the 8th International Conference on Applied Engineering (ICAE 2025)*,
Advances in Engineering Research 290,
https://doi.org/10.2991/978-94-6463-982-7_15

widespread adoption of virtual reality (VR) and augmented reality (AR) devices have further increased demand for high-quality 3D assets [3]. Three-dimensional modeling technology integrates artistic and computational principles, creating spatial depth fundamental to contemporary game development[4]

The industry's dependence on 3D assets is evidenced by the rapidly expanding digital marketplace ecosystem; platforms such as ArtStation Marketplace now offer over 98,000 high-quality game assets for both 2D and 3D development, reflecting substantial global demand from developers [5]. This expansion has catalysed advancements in various asset creation tools, including Unity 3D, C# for scripting, Blender for 3D modelling, and Substance Painter for texturing, all of which facilitate robust system development for creating engaging narrative scenes[6]. Market purchasing patterns reveal consistent trends, with May recording peak 3D game asset sales across two consecutive years among buyers ranging from major studios to independent developers [7], underscoring that 3D assets have become integral to contemporary game development workflows.

In professional game development and 3D assets, the potential and complexity of applying generative artificial intelligence (AI) in professional game development has been revealed, with the caveat that text-to-image generation has become part of the industry's main workflow, especially in pre-production phases such as ideation and prototyping [8]. The role of AI in game development is to support independent developers (indie developers) who often face resource constraints compared to large studios. This problem maximizes every investment of time and money due to limited financial support and team size. This situation demands creativity and innovation in development while facing the major challenge of producing competitive games [9]. Although indie game development costs can exceed USD 1 million, most indie developers operate on much more limited budgets [10]. This makes efficiency in 3D asset production very important. They need to create high-quality 3D assets efficiently, utilizing free 3D modeming software such as Blender.

In the technical context of 3D design, the low-poly modelling approach has emerged as a strategic solution that enables resource optimization without significantly compromising visual quality. Low-poly refers to polygon meshes in 3D computer graphics that have a relatively small number of polygons [11]. Low-poly models are faster and cheaper to produce, making them an excellent choice for indie developers or small studios with limited resources. This technique effectively optimizes resources without sacrificing essential visual quality, with polygon ranges typically between 300-50,000 triangles depending on object complexity.

3D model optimization for game development has become an important focus of academic research. A study identified 3D model optimization workflows that can enhance VR game performance and other real-time applications [12]. The research emphasized that optimization processes must consider trade-offs between visual detail and computational performance, which is crucial in game development targeting diverse platforms.

In the context of development methodology, the Game Development Life Cycle (GDLC) has been identified as an effective framework for managing game development complexity [13]. GDLC encompasses initiation, pre-production, production, testing, and release phases that can be adapted for specific 3D asset production needs. This framework enables measurable progress tracking and consistent quality assurance throughout the development process.

Research on low-poly 3D asset production in indie game development still has significant gaps in academic literature. Previous research has focused more on GDLC methods generally rather than 3D asset workflows specifically. Research by Afif, Alexandra, and Anwar examined low-poly 3D asset creation that maintains original architectural forms while optimizing polygon count for performance efficiency [14]. Setiawan and Yusuf demonstrated that low-poly assets in culture-based educational games still receive positive appreciation from players [15]. However, no research has specifically linked low-poly asset implementation with production stages in GDLC methods adapted for environment asset needs.

This research presents innovation by combining an adapted GDLC method approach with 3D Asset Production Pipeline to systematically analyse low-poly 3D asset production. The research focus is environment assets developed for 2.5D gaming contexts, with analysis based on five main indicators: workflow validation, production process efficiency, asset visual feasibility, memory performance efficiency, and asset complexity classification. This approach is expected to provide practical contributions for indie developers in optimizing 3D asset production with structured and measurable methodology.

2 Research Methods

This research employs a descriptive qualitative methodology to analyze the 3D low-poly asset production process by implementing the Game Development Life Cycle (GDLC) framework, adapted for asset production workflow and 3D Asset Production Pipeline. Six 3D low-poly environment assets developed during the IGDX Bootcamp 2024 program serve as the research subjects. These assets were selected based on their complexity levels (low, medium, high) and function as primary visual elements in the 2.5D game. Table 1 presents the detailed asset specifications.

Table 1. List of Research Assets by Complexity

No	Assets name	Complexity	Tris Estimation

1	Kiti's House	*Medium*	≤ 20.000
2	Boss Oyen's House	*High*	≤ 40.000
3	Luna's House	*High*	≤ 25.000
4	Smokey's House	*High*	≤ 30.000
5	Scratch's House	*Low*	≤ 10.000
6	Windmill	*Low*	≤ 2.000

1) GDLC Framework for 3D Asset Production

The Game Development Life Cycle (GDLC) is a process model used in game development to ensure each stage is developed systematically and structured. The new GDLC model consists of 6 development phases as shown in Figure 1.

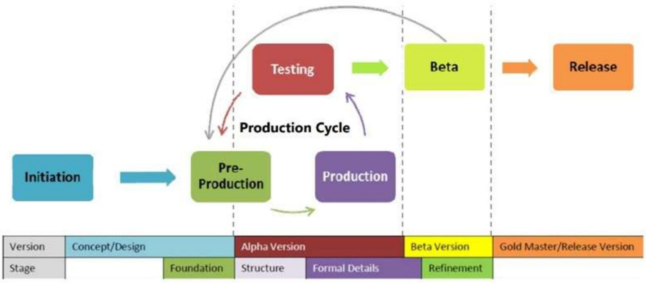

Fig. 1. Game Development Life Cycle (GDLC) Stages

Figure 1 illustrates the Game Development Life Cycle stages. GDLC or Game Development Life Cycle is a game development framework integrated with 3D Asset Production Pipeline for low-poly 3D asset production. The GDLC version used in this research is Rido Ramadhan's version, which encompasses several main stages. The process begins with the Initiation Stage, where the core game concept is formed, defining the genre, visual style specifically low-poly in this case, target audience, and fundamental gameplay. An environmental vision is established, outlining desired themes and atmosphere, often supported by reference studies. This is followed by the Pre-production Stage, which focuses on detailed planning through sketches, block out level design, and the compilation of asset lists. Key creative decisions are made here, including the color palette and mood board, alongside the establishment of technical specifications for performance optimization. The actual building occurs in the Production Stage, involving the creation of low-poly 3D assets, encompassing modeling, UV unwrapping, and texturing primarily with vertex painting. This stage also includes set dressing, lighting setup, visual effects addition, and continuous performance optimization. Next, the Testing Stage ensures quality through functionality testing (bugs), frame rate performance testing, visual consistency checks, and gameplay testing to verify the environments effectively support the game's flow. Following internal checks, the Beta Stage involves an external release for user feedback collection, which drives iterative improvements to assets and the overall environmental layout. Finally, the Release Stage is the culmination, dedicated to finalization and polishing, last-minute optimization and bug fixes, integration into the final game build, and official public distribution.

2) Data Collection Instruments

This research employs a mixed-method data analysis approach combining quantitative and qualitative analysis to provide comprehensive evaluation of Game Development Life Cycle (GDLC) implementation in 3D asset production. Quantitative analysis encompasses performance parameters such as triangle count, memory usage, draw calls, and rendering efficiency. Technical data obtained from Unity Profiler and Blender are also utilized for quantitative analysis.

The data analysis process follows Miles and Huberman's qualitative data analysis model consisting of three main stages: data reduction, data presentation, and conclusion drawing [16]. In the data reduction stage, all manually transcribed interview recordings are systematically categorized using five established evaluation benchmarks to ensure a focused and rigorous analysis. These benchmarks correlate directly with the asset development process and technical outcome. They include: production workflow validation, which assesses the efficiency and adherence to development phases; asset production efficiency, which measures time and resource utilization; visual feasibility and quality control, which checks the asset's aesthetic alignment and quality; performance efficiency and memory usage, a technical measure of optimization impact; and finally, 3D asset complexity classification, which categorizes assets based on polygon count and production demands. By grouping interview data according to these five indicators, the research facilitates a targeted analysis that links practitioner insights directly to the specific technical and creative dimensions of 3D low-poly asset creation.

This research employs data triangulation methods for cross-validation between objective software technical data and subjective expert evaluations to ensure validity and reliability of analysis results. After comparing technical objectives established in the Game Design Document (GDD) with achieved asset production results, interview transcripts with original recordings are used for validation. In the final analysis stage, assets are classified according to their complexity levels (low, medium, or high) and their influence on overall system performance. Results are then combined into recommendations for optimizing 3D asset production processes for game development.

3) Data Analysis Framework

The research evaluates 3D low-poly assets based on five primary indicators established through a triangulation methodology, covering both the process and the final output's technical and creative quality. The assessment begins with Production Workflow Validation, which measures the efficiency of the production process and its adherence to the modified GDLC (Game Development Life Cycle) phases. This is followed by Asset

Production Efficiency, a metric involving a detailed time analysis and a review of resource utilization throughout the asset creation pipeline. A critical creative evaluation is performed under Visual Feasibility and Quality Control, which verifies that the visual output meets the specific aesthetic and quality requirements for a 2.5D game. On the technical side, Performance Efficiency and Memory Usage conducts an analysis using tools like the Unity Profiler to determine the impact of asset optimization on system performance. Finally, 3D Asset Complexity Classification involves the systematic categorization of assets based on their polygon count and the overall production requirements, providing a structured approach to complexity management.

4) Expert Interview Instruments

This research employs semi-structured interviews with four sources from the game industry and visual arts. The sources were provided access to the "Village Meow" game demo before interviews for in-depth evaluation (https://igdx-meng.itch.io/village-meow). Table 2 presents The expert interview instrument was designed as Semi-Structured Interview Question Guidelines for 3D Low-Poly Asset Evaluation:

Table 2. The expert interview instrument

Question Category	Evaluation Indicators	Research Questions
Introduction	Background Expertise	1) What is your experience in 3D modeling for the gaming industry? 2) What is your opinion on the current trend of using low-poly 3D assets in games? 3) How familiar are you with the Game Development Life Cycle (GDLC) workflow?
Efficiency and Production Techniques	Production Process Optimization	1) What are the main factors that determine efficiency in low-poly asset production? 2) How do you determine the optimal number of faces, polygons, and tris?
Feasibility and Performance Evaluation	Quality Control and Visual Standards	1) What are the main indicators in assessing the visual feasibility of low-poly 3D assets? 2) How do you maintain a balance between visual detail and technical limitations?
Performance Evaluation	Memory and Performance Optimization	1) How can the effectiveness of memory reduction in created assets be evaluated?

		2) How can a balance be maintained between visual detail and technical limitations?
Suggestions and Feedback	References and Case Studies	1) What suggestions are there for improving production quality and efficiency? 2) Are there any references or case studies that can be used as a reference?

The research involved a focused series of interviews with three expert informants: a 3D lecturer and two seasoned 3D asset practitioners. These experts provided a comprehensive evaluation of the GDLC (Game Development Life Cycle) method, critically assessing both its technical implementation and the resulting efficiency and visual quality of the assets produced. The research implements cross-validation between Game Design Document (GDD) technical objectives and actual production outcomes. Expert interview transcripts are validated against original recordings to ensure data integrity. Assets undergo systematic classification based on complexity levels and their impact on overall system performance, providing comprehensive evaluation framework for optimization recommendations. This section presents the results of 3D low-poly asset production analysis based on technical evaluation using Unity Profiler, Blender Analytics, and expert validation through structured questionnaires. The analysis focuses on technical aspects of asset creation, production efficiency, and resulting visual quality.

3 Result and Discussion

A. Initiation Stage

The Initiation stage is crucial as it lays the foundation for successful development by comprehensively establishing the game concept. During this foundational stage, key elements were successfully determined: the game genre was set as a 2.5D life simulation adventure game; The target audience is defined as adolescence [17] to young adults [18] in the 13-30 age range. who appreciate cozy and narrative-driven games; and the core gameplay mechanics were solidified around exploration and social interaction between cat characters, forming the essential basis for the subsequent development. The environmental vision established adopts a cat village theme with Indonesian cultural nuances, creating a warm, peaceful, and comfortable (cozy) atmosphere while harboring hidden tension due to the influence of Boss Oyen's tyranny. The Game Environment Concept description is presented in Table 3.

Table 3. Game Environment Concept Description

No	Aspects	Description
1	Title of Game	Village Meow
2	Genre	Life Simulation Adventure with narrative elements
3	Visual Style	Low-Poly 3D Environment with 2D characters (2.5D)
4	Target Audience	Adolescence to young adults (13–30 years old), fans of cozy, narrative, and relaxing games PC
5	Platfrom	PC
6	Theme	Semi-Fantasy with a traditional Indonesian village feel
7	Gameplay Focus	Exploration, Social Interaction, and Light Resource Management
8	Technical Target	Stable 60 FPS, total polycount per main asset under 5,000 tris, efficient draw calls, textures sized up to 512 x 512 for optimization mobile.

The reference study was extensive, generating over 20 inspiration sources from games known for their effective low-poly styles, such as Paper Mario, Born of Bread, and Alto's Adventure. This competitor analysis clearly demonstrated that a low-poly approach is highly effective for crafting engaging visual experiences without compromising performance. Several key insights emerged from this analysis: low-poly aesthetics are adept at conveying narrative themes while maintaining technical efficiency; Indonesian cultural elements can be seamlessly integrated into low-poly environmental designs; and finally, the cozy game genre particularly benefits from simplified geometric forms, which serve to emphasize atmosphere rather than intricate detail.

The initiation stage successfully established a cohesive creative direction that balances artistic vision with technical constraints. The Indonesian cat village concept provides a unique cultural context while the low-poly approach ensures scalability for independent development resources. This foundation enables systematic asset production aligned with both aesthetic goals and performance requirements.

B. Pre-Production Stage

During the pre-production stage, detailed planning was successfully completed, serving as a reference for 3D asset creation in the development of low-poly 3D house characters. Character design was developed through 3 iterations of sketches and concept art to test the personality and visual appeal of house characters.

1) House Character Design and Proportions

The house character design process involved a systematic iterative development approach, aimed at establishing optimal visual characteristics for the 2.5D game environment. Initial concept exploration concentrated on finding a balance between architectural authenticity and character anthropomorphization; this ensured that the structures retained recognizable building features while simultaneously incorporating personality traits suitable for narrative interaction. The design iteration process unfolded in three key stages: the First Iteration established basic architectural forms with minimal character features; the Second Iteration focused on enhancing personality elements while carefully maintaining structural integrity; and the Third Iteration achieved an optimized balance between overall character appeal and the specific technical requirements of a low-poly aesthetic.

The proportional system developed during this stage established standardized scaling relationships between different house types, ensuring visual coherence across the game environment while accommodating varying complexity levels for performance optimization. Figure 2 shows the height chart for house characters. During the concept design process, house character designs were created with heights ranging from 100 cm to 240 cm (cat world scale), with proportions adjusted according to the characteristics of each inhabitant. House components such as eyes, ears, and roof colors were designed to reflect character characteristics and support visual storytelling flow in the Village Meow world.

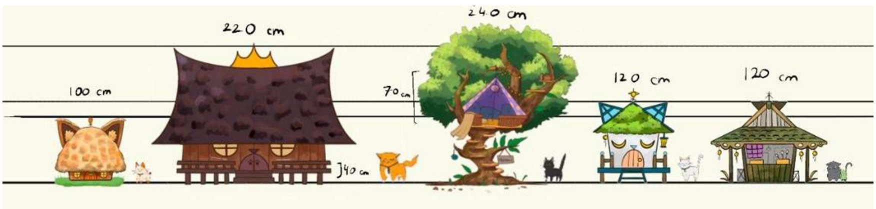

Fig. 2. *The Height Chart for House Characters*

2) Character Art Direction

Fig. 3. *Mood board and Reference Environment*

The mood board and color palette shown in Figure 3 were established using a combination of warm yellow, moss green, wood brown, and pastel blue that creates a cozy, cheerful, and friendly personality[19]. Technical specifications were set with a target maximum of 800 polycount per individual component, and total character polycount not exceeding 3000 to maintain optimal performance.

Fig. 4 Mood Board and Kiti's House References

Figure 4 shows the mood board and visual references for Kiti's House design. This house is inspired by traditional Indonesian house forms, with visual elements resembling cat characteristics, such as roof shapes resembling ears.

C. Production Stage

The Production stage represents the execution phase of asset production through a 3D Asset Production Pipeline integrated with low-poly modeling principles [20]. The production pipeline was designed to optimize workflow from modeling to final asset delivery while considering game performance requirements.

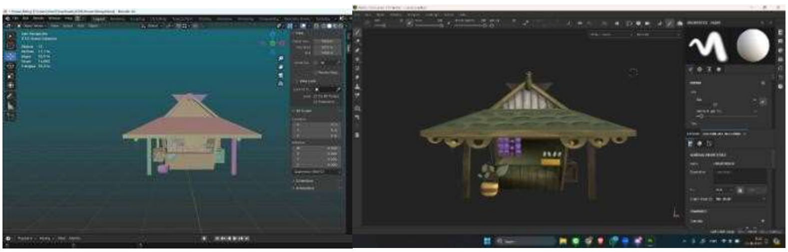

Fig. 5. The 3D modeling and texturing stage of Smokey's House using Blender.

Figure 5 shows the 3D modeling stage of Smokey House using Blender. The process begins with a low mesh base that is gradually enhanced. The texturing process contains UV Unwrapping and Texturing stage using Adobe Substance Painter with hand-painted workflow implementation that not only provides unique artistic character but also maintains file size efficiency critical for game performance.

D. Testing Stage

The Testing stage represents a comprehensive evaluation phase where every technical and visual aspect of low-poly 3D house assets is systematically validated. The testing process includes rendering performance testing, hardware compatibility, and visual quality assessment under various lighting conditions. The integration testing process involves technical validation through Unity Profiler to measure real performance metrics such as triangle count, memory usage, draw calls, and rendering effects, which can be compared with initial target specifications.

Technical performance evaluation in this research uses two main approaches to ensure accuracy and comprehensiveness of measurement. Unity Profiler Analysis was implemented as the primary instrument for real-time performance analysis. Technical evaluation was conducted on six low-poly 3D assets produced in this research. Each asset was evaluated based on critical technical parameters including triangle count, vertex count, file size, memory usage, UV coverage, and geometric complexity level. Table 4 shows the technical evaluation results conducted on the 6 low-poly 3D assets produced and analyzed in this research.

Tabel 4. The technical evaluation results conducted on the 6 low-poly 3D assets

Nama Aset	***Triangel Count***	***Vertex Count***	***File Size (MB)***	***Memory Usage (GB)***	***Draw Calls***	**Kom-pleksitas**
Kiti's House	25.932	13.008	8.41	0.68	*4*	*Medium*
Boss Oyen's House	136.836	68.113	22	0.67	*16*	*High*
Luna's House	20.942	10.748	3.97	1.22	*59*	*High*
Smokey's House	41.250	21.026	4.58	0.65	*4*	*High*
Scratch's House	7.061	3.503	3.69	1.20	*22*	*Low*
Windmill	1.288	692	84.2	0.5	*5*	*Low*

Technical data shows significant variation in asset complexity, with triangle count ranging from 1,288 (Windmill) to 136,836 (Boss Oyen House). File size varies between

3.69 MB to 84.2 MB, while memory usage ranges from 0.5 GB to 1.22 GB. This variation reflects an approach tailored to the function and visual importance of each asset in the game environment context.

E. Specific Findings

Complexity classification was based on triangle count, where assets with less than 10,000 triangles were categorized as low complexity, 10,000-30,000 triangles as medium complexity, and more than 25,000 triangles as high complexity. This classification refers to industry standards for game development targeting mobile platforms and entry-level desktop that require strict performance optimization.

Draw calls analysis revealed an interesting pattern where Luna's House has the highest number of draw calls (59), followed by Scratch's House (22), despite both having relatively low geometric complexity. This indicates the use of more complex and diverse materials or textures on both assets. However, Kiti's House and Smokey's House show good efficiency with only 4 draw calls each, indicating better material optimization.

Memory usage shows patterns that do not always correlate with geometric complexity. Luna's House and Scratch's House have high memory usage 1.22 GB and 1.20 GB despite having relatively low triangle counts, indicating the use of high-resolution textures or more complex materials. However, Boss Oyen's House with the highest geometric complexity only uses 0.67 GB memory, it shows very good optimization.

Overall, the asset complexity distribution shows diverse design strategies, with 50% assets categorized as high complexity, 16.7% as medium complexity, and 33.3% as low complexity, reflecting the balance between visual quality and performance optimization for various gameplay needs.

A. **Kiti's House (Medium Complexity)**
   The creation process of Kiti's House began with a block-out modeling approach using basic primitive shapes. Box modeling technique was applied with extrusion and loop cuts to form the basic building structure.
B. **Boss Oyen's House (High Complexity)**
   Boss Oyen House uses an architectural modeling approach with focus on characteristic details reflecting character personality. The modeling process uses a modular approach with reusable components such as window frames, door elements, and decorative features.
C. **Luna's House (High Complexity)**
   Luna's House production uses organic modeling techniques with subdivision surface modifier to create curved surfaces. Texturing workflow uses Substance Painter with hand-painted base layers and procedural details for weathering effects.
D. **Smokey's House (High Complexity)**

Smokey's House uses a modular construction approach with reusable components. Boolean operations are used strategically to create cutting details on building structures.

E. Scratch's House (Low Complexity)
Scratch's House uses a minimalist modeling approach with extensive geometric simplification. Texturing uses a flat color approach with minimal shading complexity. UV layout is optimized with a single texture sheet to reduce draw calls.

F. Windmill (Low Complexity)
Windmill uses radial symmetry modeling with a modifier-based approach. Array modifier and mirror modifier are used to create consistent blade geometry.

The results of the qualitative descriptive analysis of expert feedback are shown in Table 5.

Table 5. Results of Comparative Analysis of Expert Feedback

Assets	Amirul Mu'minin (lecturer/ Practitioner)	Muhammad Ilham (Lead 3D Artist)	I Made Wahyu (3D Artist)	Recommendation Consensus
Kiti's House	It is best to make it a single asset. Do not make it too detailed on the inside because it will not be visible.	Hand-painted texturing is already wise in choosing efficiency techniques and art style.	Visually it's okay, but more detailed topology checking is needed to ensure tris efficiency.	**Optimization:** Remove hidden geometry, implement modular design, review detailed topology
Boss Oyen's House	Triangle count 8000+ is too high for low-poly. It must be reduced to a maximum of 5000. Decimation is required in Blender.	Detailed checking of the topology is necessary because efficiency cannot be confirmed without seeing the wireframe.	A Unity profiler check is needed to see the actual impact on performance.	**Critical:** Review with Unity Profiler, urgent retopology, optimization based on actual performance data
Luna's House	The back part that is not visible can be removed.	Visually, it's fine, but tris count documentation is needed to ensure efficiency.	The art style is in line with the low-poly concept, but it is necessary to ensure draw calls.	**Good:** Optimized geometry, image call management, maintaining artistic style consistency.

Smokey's House	There are no significant problems, as long as the triangle count remains within the maximum limit of 5000.	It is necessary to evaluate whether there are any unused assets in the scene.	It is necessary to ensure that the asset workflow can be reimported properly for team collaboration without damaging the scene arrangement.	**Accepted:** Monitoring the number of triangles, implementing workflows that support team collaboration
Scratch's House	It is already visually and performance efficient.	Example of a good workflow.	Most efficient in terms of art style and technical aspects.	**Excellent:** Benchmark for other low-complexity assets as templates

According to the comparative analysis of expert feedback shown in Table 5, the common strengths identified include a visually appealing and consistent low-poly art style, as well as the successful optimization of a few assets that serve as a "Excellent" benchmark. The most common optimization issues are critical issues with asset geometry, specifically excessive triangle counts (more than 5000 maximum limit) and a lack of systematic performance testing or topology documentation. As a result, the priority recommendations call for an immediate retopology (reduction of hidden and excessive geometry), a mandatory review with the Unity Profiler, and the implementation of strict workflows to ensure both efficiency and team collaboration support.

F. Beta and Release Stage

The Beta stage represents an integration testing phase where 3D house assets are tested in actual gameplay context involving end-users or beta testers. Assets are evaluated from player experience perspective and visual harmony with other environments. Figure 6 shows the internal bug process where the team conducts internal testing to identify and fix problems found in the game.

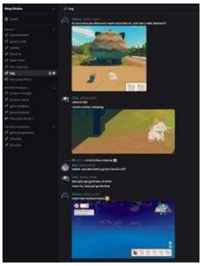

Fig.6. Internal testing to identify game's bug

G. Release Stage

The Release stage represents the finalization phase where low-poly 3D house assets have reached production-ready status and are ready for deployment to end-users. This process involves final packaging, platform-specific optimization, and preparation for distribution through various channels. Figure 7 shows Village Meow Game published as a demo on the itch.io platform as a strategic approach for market validation and community engagement.

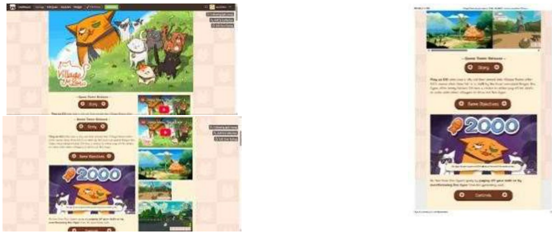

Fig.7. Release a demo of Village Meow Game on itch.io

4 Conclusion

Based on the conducted research, it can be concluded that the implementation of Game Development Life Cycle (GDLC) framework integrated with 3D Asset Production Pipeline proves effective in optimizing low-poly 3D asset production for indie game development. Expert feedback evaluation confirms that the applied hand-painted texturing technique is appropriate for low-poly art style and provides good efficiency in resource usage. These findings provide practical contributions to the indie developer community in optimizing 3D asset production with structured and measurable methodology.

Research results show that the implementation of GDLC framework adapted for 3D asset production demonstrates positive results, aligned with findings by Hristov et al. [21] who emphasize the importance of structured workflow in game asset development. Clear stages from initiation to release provide quality gates that enable consistent quality control. However, this research found that GDLC adaptation requires specific modifications to accommodate the specific needs of 3D asset production, especially in the production stage that requires more detailed sub-phases for modeling, texturing, and optimization.

References

[1]. Markets and Markets, "3D Digital Asset Market - Global Forecast to 2029," Online 2024. https://www.marketsandmarkets.com/Market-Reports/3d-digital-asset-market-109920847.html (accessed June 24, 2025).

[2]. Research Nester, "3D Gaming Technology Market - Historic Data (2019-2024), Global Trends 2025, Growth Forecasts 2037," Online 2025. https://www.researchnester.com/reports/3d-gaming-technology-market/6876 (accessed June 24, 2025).

[3]. Grand View Research, "3D Digital Asset Market Size, Share & Trends Report, 2030," Online 2024. https://www.grandviewresearch.com/industry-analysis/3d-digital-asset-market-report (accessed June 24, 2025).

[4]. Y. Lan, "Development and Application of 3D Modeling in Game," *Academic Journal of Science and Technology*, vol. 7, no. 2, pp. 94–97, Sep. 2023, doi: 10.54097/ajst.v7i2.11949.

[5]. ArtStation, "3D Assets - Game Development," *ArtStation Marketplace* n.d. https://www.artstation.com/marketplace/game-dev/assets/3d (accessed June 24, 2025).

[6]. R. Kaźmierczak, R. Skowroński, C. Kowalczyk, and G. Grunwald, "Creating Interactive Scenes in 3D Educational Games: Using Narrative and Technology to Explore History and Culture," *Applied Sciences*, vol. 14, no. 11, p. 4795, Jun. 2024, doi: 10.3390/app14114795.

[7]. CGTrader, "What Game Developers Are Buying Right Now: Trends in 3D Asset Demand," *CGTrader Blog* 2025. https://www.cgtrader.com/blog/what-game-developers-are-buying-right-now-trends-in-3d-asset-demand (accessed June 24, 2025).

[8]. Veera Vimpari, Annakaisa Kultima, Perttu Hämäläinen, and C. Guckelsberger, "'An Adapt-or-Die Type of Situation': Perception, Adoption, and Use of Text-to-Image-Generation AI by Game Industry Professionals," Proceedings of the ACM on human-computer interaction, vol. 7, no. CHI PLAY, pp. 131–164, Sep. 2023, doi: https://doi.org/10.1145/3611025.

[9]. HowToGeek, "What Indie Game Developers Want You To Know in 2024," 2024. https://www.howtogeek.com/what-indie-game-developers-want-you-to-know-in-2024/ (accessed June 24, 2025).

[10]. Ediiie, "Budgeting Your Indie Game in 2024: Development Cost Insights," 2024. https://www.ediiie.com/blog/indie-game-development-cost/ (accessed June 24, 2025).

[11]. Keleşoğlu MM, Özer DG, "A Study on Digital Low Poly Modeling Methods as an Abstraction Tool in Design Processes," *Civil Engineering and Architecture*, Vol. 9(7), 2021.

[12]. Silva PP e, Ruas VHC, Oliveira GCD, Martinelli TF, de Oliveira AVM, Schimidt MQ, et al., "Workflow to Optimization of 3D Models for Game Development," 2018 *20th Symposium on Virtual and Augmented Reality (SVR), IEEE*, p. 225–9, 2019. https://doi.org/10.1109/SVR.2018.00041.

[13]. Ramadan R, Widyani Y, "Game development life cycle guidelines," 2013 *International Conference on Advanced Computer Science and Information Systems (ICACSIS), IEEE*, p. 95–100, 2013. https://doi.org/10.1109/ICACSIS.2013.6761558.

[14]. Afif RT, Alexandra QM, Anwar AA, "Perancangan 3d Modelling Environment Low Poly Rumah Adat Kampung Naga Kabupaten Tasikmalaya," *Jurnal PATRA* vol 6, 2024.

[15]. Fitriyani Asiqin GR, Sumaryana Y, Hidayat CR, "Game Edukasi Pembelajaran Budaya Flores Berbasis Android Dengan Menggunakan Metode Game Develoment Life Cylce (GDLC)," *Jurnal Informatika Dan Teknik Elektro Terapan*, Vol 13, 2025. https://doi.org/10.23960/jitet.v13i1.5917.

[16]. Miles MB, Huberman AM, "Analisis data kualitatif: buku sumber tentang metode-metode baru," *Jakarta: UI-Press*, 1992.
[17]. S. M. Sawyer, P. S. Azzopardi, D. Wickremarathne, and G. C. Patton, "The age of adolescence," *Lancet Child Adolesc Health*, vol. 2, no. 3, pp. 223–228, Mar. 2018, doi: 10.1016/S2352-4642(18)30022-1.
[18]. J. J. Arnett, "Emerging adulthood: A theory of development from the late teens through the twenties.," *American Psychologist*, vol. 55, no. 5, pp. 469–480, 2000, doi: 10.1037/0003-066X.55.5.469.
[19]. R. Chen, "The Effect of Color on People's Emotions," Journal of Social Science Humanities and Literature, vol. 7, no. 2, pp. 175–178, Apr. 2024, doi: 10.53469/jsshl.2024.07(02).21.
[20]. Perez E, "The design of indie games, a different paradigm," *Thesis for Doctoral* 2019.
[21]. Hristov G, Kinaneva D. A, "Workflow for Developing Game Assets for Video Games," 2021 *3rd International Congress on Human-Computer Interaction, Optimization and Robotic Applications (HORA), IEEE*, p. 1–5, 2021. https://doi.org/10.1109/HORA52670.2021.9461355.